

Is Air Pollution Regulation Too Lenient? Evidence from US Offset Markets[†]

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We develop a framework to estimate the marginal cost of air pollution regulation and apply it to assess policy efficiency. We exploit a provision of the Clean Air Act that requires new plants to pay incumbent facilities to reduce emissions. This “offset” policy creates hundreds of local pollution markets, differing by pollutant and location. Theory and transaction data suggest that offset prices reveal marginal abatement costs. We compare these prices to marginal benefits of pollution reduction estimated using leading air quality models and find that, on average, marginal benefits exceed marginal costs by more than a factor of ten. (JEL D61, H23, K32, Q52, Q53, Q58)

A classic idea in economics is that firms may provide too much of an externality like pollution because they do not account for its full social costs. Policies addressing this market failure can maximize social welfare by regulating emissions until the marginal cost of complying with policy equals the marginal social benefit of reducing pollution emissions (Pigou 1932). In practice, designing policies to limit negative externalities involves a delicate balancing act between the costs to firms of compliance and the benefits to society of reducing externalities. For externalities including crime, innovation, and smoking, it can be difficult to estimate the total social costs and benefits of a policy, let alone the marginal costs and marginal benefits. Thus, it can be challenging to know whether existing policies maximize social welfare or are more or less stringent than economic efficiency requires.

These issues are particularly important for air pollution. Some research estimates that over 5 percent of premature US mortality comes from air pollution, and over

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a third of measured benefits of all recent federal regulations came from reducing one air pollutant, particulate matter (Fann et al. 2012; Dominici, Greenstone, and Sunstein 2014). Cleaning up air pollution may be costly—US air quality for some pollutants has improved by over 90 percent since 1970 (Currie and Walker 2019; Shapiro 2022). The marginal cost of pollution abatement typically increases with the quantity of abatement, so these enormous pollution decreases lead to the question of when the marginal costs of increasing regulation begin to exceed its marginal benefits.

Political and economic debates have echoed these textbook questions. Two law professors, for example, summarized, “[The Environmental Protection Agency’s (EPA’s)] ozone standard is insufficiently stringent, not overly expensive” (Livermore and Revesz 2015). Oates (1997) disagreed:

Existing benefit-cost studies suggest that total benefits for air pollution control have substantially exceeded the total costs in the United States, but this, of course, does not really address the question. It is still possible, and I think likely in view of the evidence, that for several of the criteria air pollutants (such as ground-level ozone), the standards have been pushed well beyond the point where the marginal benefits equal marginal cost (pp. 1327-8).

Government and researchers obtain varying results, for example, on whether the benefits of tightening ozone pollution standards exceed their costs (Office of Management and Budget 2009; Lange, Mulholland, and Honeycutt 2018).

In part to help reconcile these disparate views, we develop a framework to estimate the marginal cost of pollution abatement separately by pollutant, county, and year. We utilize an important yet underexplored provision of the Clean Air Act that forbids increases in pollution emissions from large industrial sources in areas where pollution exceeds the National Ambient Air Quality Standards (NAAQS). Large polluting plants entering these “nonattainment areas” must offset their potential emissions by paying an incumbent polluter in the same area to permanently reduce their emissions. Entrants can spend millions of dollars purchasing such pollution “offsets.” Market participants describe spending on these offsets as among the largest environmental expenditures for new or expanding facilities.

We analyze newly available offset transaction records from two states that require public disclosure of contracts, and we purchased additional data on fourteen other states plus Washington, DC, from a leading firm that advises offset transactions. Most of these data have never been analyzed or discussed in government or academia. Our data cover over 40 markets and 60 percent of economic activity from US offset trading areas. Our analysis also reflects extensive interviews with market participants, including regulators at the EPA, the California Air Resources Board, and local air districts; brokers and traders active in offset markets; and environmental managers at firms that have traded offsets.¹

Offset markets provide information about the marginal cost of air pollution regulation. A new entrant must purchase enough offsets to cover their emissions. When

¹ Additionally, we repeatedly attempted to contact every person listed as involved in trading offsets between 2018 and 2020 in Houston, the market where government documents provide traders’ contact information.

the market price for offsets exceeds an entrant's marginal cost of abatement, the entrant should invest in pollution abatement until its cost of abatement approaches the market price for offsets. As abatement costs begin exceeding offset prices, a profit-maximizing firm should purchase offsets for the remaining emissions. Incumbents also have an incentive to generate and sell offsets when the market price for offsets exceeds marginal abatement costs. Incumbents may generate offsets by investing in permanent, verifiable emissions reductions, which regulators certify. Canonical views of environmental markets dating back to Dales (1968) and Crocker (1966) and formalized by Montgomery (1972) suggest that emissions markets reveal marginal abatement costs via allowance prices and that efficient environmental policy equates the price of pollution offsets to the marginal social benefits of pollution abatement. The marginal benefits of abatement include health and other benefits from emissions reductions. These benefits differ by pollutant and location.²

We begin with several stylized facts on offset markets. We show that offset markets cover 60 percent of US population or economic activity, representing 180 million people or \$11 trillion in GDP. Offset prices are rapidly rising in real terms. Offset prices are related to proxies for regulatory stringency, including pollution abatement costs, emissions, economic activity in polluting industries, and nonattainment severity.

We then test policy efficiency. In markets where offset prices exceed the marginal benefits of abatement, we conclude that air quality regulation is more stringent than is efficient. In markets where the marginal benefits of abatement exceed mean offset prices, regulation is more lenient than is efficient.

We find that for most pollutants and markets with data, regulation is much less stringent than efficiency would require, though recent regulations in Houston are an exception. Nationally, mean marginal benefits of abatement for nitrogen oxides (NO_x) and volatile organic compounds (VOCs) are ten times mean offset prices. We find large benefit/cost ratios in each US region in most markets for one important other pollutant with limited data (particulate matter) and in sensitivity analyses. In Houston, however, the marginal benefits of abating VOCs are only half of mean offset prices.

We provide the first comprehensive empirical analysis of US air pollution offset markets. A substantial body of research analyzes the Clean Air Act. Given the importance of offset markets, it is noteworthy that empirical research has largely overlooked them. Among market-based environmental policies, economists once described Clean Air Act offsets and related policies as by far the most important of these programs in terms of scope and impact (Cropper and Oates 1992), but the limited existing research mentioning offset markets mostly describes legal or policy details (Dudek and Palmisano 1988; Stavins 2003; National Research Council 2006; Leonard 2018).

We also provide revealed preference estimates of the marginal cost of abatement that differ by pollutant, location, and year; we then compare these to corresponding marginal benefits. Since the benefits of emissions reductions differ by pollutant and location, it is important for analysis of efficiency to have cost estimates that also differ by pollutant and location. Put another way, optimal regulation of externalities is more stringent in areas where marginal damages are greater.

²Externalities from criteria air pollutants, like those the Clean Air Act regulates, are fairly local and larger in more densely populated areas.

To date, however, the literature has essentially no revealed preference estimates of marginal abatement costs that differ by pollutant and location.³ Existing estimates of marginal abatement costs typically come from estimating structural cost functions of one industry, for example, sulfur dioxide abatement from electricity generation (Gollop and Roberts 1985; Carlson et al. 2000). These methods require a thorough understanding of costs and production constraints and information on input prices and quantities. These methods are not ideal in settings where emissions come from many industries. In isolated settings, researchers do observe permit prices in cap-and-trade markets as a measure of abatement costs (Fowlie, Knittel, and Wolfram 2012; Deschênes, Greenstone, and Shapiro 2017). These markets are few, geographically large, and preclude identifying costs separately for different pollutants or locations within a market.

Our focus on marginal abatement costs is useful since economic research has focused more on measuring the marginal benefits than marginal costs of air pollution policy.⁴ We also contribute to an emerging literature quantifying how regulatory stringency evolves over time. Other work estimates trends in the shadow price of pollution, using models requiring strong assumptions (van Soest, List, and Jeppesen 2006; Shapiro and Walker 2018). Tracking offset prices over time helps characterize trends in regulatory stringency nationally and by market.

Two clarifications may be useful. First, many environmental regulations govern air and water pollution, and one might wonder how these regulations affect our interpretations of the efficiency of the Clean Air Act's NAAQS. While other environmental policies create costs for firms, they do not change a plant's marginal decision—increasing expenditures on abatement decreases required expenditures on offsets. An optimizing plant should equate these costs at the margin.

Second, what can we learn from studying these markets, which can be thin? While we only analyze entrants and incumbents trading offsets, these represent the marginal emissions decisions. Although not every incumbent trades offsets, because every incumbent can, these marginal costs provide information about all incumbents and not merely about those that trade offsets.⁵ We analyze thousands of transactions across 9 to 25 years per market for two common pollutants, covering over 40 markets, including the seven largest US metro areas. We consider four separate air quality and valuation models, two quasi-experimental and four epidemiological estimates of the main elasticity determining benefits (the mortality-concentration response function for particulate matter), and three different estimates of the value of a statistical life (VSL). While one market or specification may be unrepresentative, and while we show important heterogeneity across markets, these estimates provide a broad picture of regulatory efficiency for the country's largest offset markets.

³In theory, engineering models used by the EPA can provide estimates of marginal abatement costs. The data and informational requirements, however, far exceed the administrative and resource capacity available. Moreover, engineering estimates do not use revealed preference, typically ignore nonpecuniary abatement costs, and can differ substantially from revealed preference estimates (Carlson et al. 2000; Fowlie, Greenstone, and Wolfram 2018; Allcott and Greenstone 2023).

⁴For example, in five top general-interest economics journals, our review found 13 papers on the marginal benefits of air pollution regulation and only three on the marginal costs (Appendix A). The three cost studies analyzed focused settings—sulfur dioxide emissions from coal-fired power plants or vehicle environmental inspections in developing countries.

⁵Similarly, plants in cap-and-trade markets may hold allowances without trading them.

I. History and Design of US Air Pollution Offset Markets

The following descriptions draw on existing work (Dudek and Palmisano 1988; Fraas, Graham, and Holmstead 2017; Leonard 2018).⁶

A. Offset Market History

The 1970 Clean Air Act Amendments directed the EPA to set standards for ground-level ozone, particulate matter, and four other “criteria” air pollutants. The EPA monitors compliance with these standards by operating air quality monitors. If readings from a monitor violate the pollutant standard, the county enters “nonattainment,” and polluting firms face stricter regulation.

Since the 1977 Clean Air Act Amendments, new or expanding facilities have been subject to New Source Review, which prohibits *net* increases in pollution emissions from stationary sources (i.e., facilities) in nonattainment areas. This provision allowed large polluting plants to enter nonattainment areas if they offset the resulting increase in pollution emissions with decreases in emissions of the same pollutants from incumbents in the same area. Hahn and Hester (1989) discuss early literature on emissions offsets and offset markets in the Clean Air Act; Supplemental Appendix B provides additional background.

B. Offset Market Design

An incumbent in a nonattainment area may choose to permanently decrease its emissions and then receive an offset from regulators specifying the tons per year, pollutant, and nonattainment area. Installing more stringent abatement technology than required, closing down a plant or part of it, and decreasing total production can all generate offsets.

To generate an offset, the EPA requires that emissions reductions must be permanent, quantifiable, federally enforceable, and surplus. Surplus means the reduction is not required by current regulations, relied on for state implementation plan planning purposes, and not used to meet any other regulatory requirement.” Federally enforceable means the “reduction is enforceable through rule or permit (US Environmental Protection Agency 1980). Offsets and a plant’s air quality permit may specify the maximum permitted hours of operation, production rate, or input rate, and ways to guarantee compliance (Gauna 1996). Quantifiable means the actual emissions reduced are able to be calculated.⁷ Permanent means the “reduction is unending or indefinite,” which often requires installing or removing capital equipment, documented in photographs and records (Rucker 2018). Some offset markets, particularly for carbon dioxide, suffer from concerns that offsets are not surplus or additional. Such concerns have not been prominent for the markets we study where states and the EPA tightly enforce requirements.

⁶Formally, offsets are called “Emission Reduction Credits.” We refer to sources that are new or undergoing significant modifications as “entrants.” We refer to stationary pollution sources as “plants.”

⁷Quantification may use “source tests, mass balances, or other means acceptable to the permitting agency” (Leonard 1996, p. 81). Generally, analysis must use the same method to quantify baseline and postintervention emissions (Haddad and Palmisano 2001).

An incumbent can sell an offset to entrants, possibly in parts. Regulators typically maintain a registry of offsets and certify transactions. Offset markets are nonetheless decentralized, so buyers and sellers write bilateral offset contracts.

Some offset transactions involve brokers. Brokers help entrants find inexpensive offsets, including offsets in the process of being generated or that could be generated in the future, and provide expertise in regulation and environmental contracting. Brokers also provide confidentiality—because transactions can be a public record, a plant that trades offsets without a broker is publicly signaling its expansion or contraction plans, which may reveal sensitive business information.

Offsets are documented in the purchaser's air quality permit, which is generally required for a polluting plant to start construction or operate. Precise rules governing offsets differ by state or air district. Supplemental Appendix B.3 discusses one example offset.

What determines offset prices? The demand for offsets reflects the demand for entry or expansion of polluting firms and associated abatement costs. Offset supply reflects the availability of abatement opportunities. A few examples from market participants illustrate how supply and demand help explain offset price variability within and across markets. Los Angeles has been in nonattainment since the 1970s. Plant entry has exhausted most inexpensive abatement opportunities from incumbents. Thus, the limited *supply* of offsets produces Los Angeles' high offset prices. Conversely, in the 2010s, Houston experienced a large increase in *demand* for offsets to accommodate new refining capacity. Demand for new refining capacity stemmed from the abundance of cheap natural gas discovered in west Texas due to hydraulic fracturing. This demand outstripped supply, and offset prices rose. The Upper Green River Basin in Wyoming recently entered nonattainment and is an area with limited demand for building large polluting plants, so it has low offset demand. This area also has incumbents with inexpensive abatement opportunities and low offset prices.

II. Data

This section describes data; Supplemental Appendix C provides additional details.⁸

A. Offset Markets

We obtain offset data from several sources. For 14 states plus Washington, DC, we use records describing NO_x and VOC offset transactions from a leading emissions offset brokerage and advisor, Emission Advisors (Emission Advisors Inc. 2020).⁹ These records list mean prices in each market $\times$ year and describe market size in four bins. We also use transaction-level records from the California Air Resources Board over 1993–2018 and the Texas Commission on Environmental Quality over

⁸Beyond the data discussed in this section, the analysis uses a number of additional datasets discussed in Supplemental Appendix C: Arias (2014); Centers for Disease Control and Prevention (2020a,b); Federal Reserve Bank of St. Louis (2020a,b); National Bureau of Economic Research (2020a,b); Shapiro and Walker (2025); US Bureau of Labor Statistics (2023); US Census Bureau (2017, 2019, 2020a,b,c); and US Environmental Protection Agency (2019a, b).

⁹The 14 states are Arizona, Connecticut, Delaware, Illinois, Indiana, Maryland, Missouri, New Jersey, New York, Ohio, Pennsylvania, Virginia, Wisconsin, and Wyoming.

2001–2019 (California Air Resources Board 2019, 2020; Texas Commission on Environmental Quality 2019a, b). The California data list price, quantity, and the associated air quality district. The Texas data also list names of the selling and buying firms and an identifier code tracking the offset's lifecycle. We exclude intrafirm and temporary offset transactions.

Most offsets represent the permanent right to emit a ton of pollution. Most estimates of the marginal benefits of abating criteria pollution represent the marginal benefits of decreasing emissions of one ton of pollution in one year. To compare permanent offset prices and temporary pollution abatement benefits, we infer what price offset transactions would have had if the offsets only lasted one year. Supplemental Appendix B.4 describes the process of converting permanent offset prices to annual prices using the ratio of short-term to permanent offset transactions. Supplemental Appendix C.4 discusses how the price ratios between temporary and permanent offsets imply a discount rate for these assets of 8–10 percent.

B. Marginal Benefits of Pollution Abatement

We use estimates of the marginal benefits of pollution abatement from a leading “integrated assessment” model, AP3. Much research uses AP3 and its predecessors (e.g., Muller and Mendelsohn 2009; Fowlie, Greenstone, and Wolfram 2018). Using AP3, we calculate the benefit of a one-ton decrease in pollution emissions for each county and pollutant. We use a modified version of AP3 code that Nick Muller generously provided (see Supplemental Appendix C.3.2) (Muller 2020); this change increases the marginal benefits of abatement by 7.5 percent. Appendix Figure 1 maps the marginal benefits of pollution abatement from AP3, which vary by pollutant and county. The marginal benefits of abatement increase with population density.

We report several sensitivity analyses. Our baseline estimates use the EPA's preferred VSL of \$8.8 million (2017 dollars) (US Environmental Protection Agency 2010). We consider one alternative estimate of \$3.7 million from the OECD (2012) and an age-adjusted VSL (Murphy and Topel 2006; Carleton et al. 2022). Our baseline estimate of the PM_{2.5} concentration-adult mortality response function, which accounts for most estimated air pollution damages, is from Krewski et al. (2009). We report alternatives from the fifth and ninetieth percentile of the confidence interval from Krewski et al. (2009), from another epidemiological study (Lepeule et al. 2012), from a cross-sectional regression discontinuity estimate from China (Ebenstein et al. 2017), and from a panel data estimate using nonattainment designations as an instrument for pollution (Sanders et al. 2020). We also assess sensitivity to three other air quality and valuation models—InMAP, EASIUR, and AP2.

Supplemental Appendix E describes additional data used for sensitivity analyses that explore the relationship between offset prices and existing measures of regulatory stringency.

III. Describing Offset Markets

We focus on the mean offset price in each market $\times$ year for several reasons. The mean is more robust to outliers than the maximum, especially given the possibility

of reporting errors, measurement error, or idiosyncratic events disproportionately impacting one offset transaction. If offset prices gradually increase as we move up the marginal abatement cost curve, using mean prices helps reflect these forces. Additionally, the data we purchased on 14 states plus Washington, DC only report the mean, so we can only analyze the median and maximum price for California and Texas. We do report sensitivity to the median and maximum price for key results in the markets for which these data exist.

A. Offset Market Coverage

Appendix Table 1 describes US air pollution offset markets and the markets in our data. Panel A shows that the United States has several hundred offset markets that cover areas with 180 million people or \$11 trillion in GDP; this represents around 60 percent of the US population, GDP, or manufacturing employment (Bureau of Economic Analysis 2020; US Bureau of Labor Statistics 2020b). Market sizes are skewed; a large share of markets represents low-population areas with few transactions. Panel B shows that during the years of our study the United States had 226 offset markets covering 50 percent of the US population or 60 percent of economic activity. Panel C shows that our data cover 42 markets of these 226, representing 60 percent of the total population share in offset markets during this time or 63 percent of economic activity in offset markets.

Most reviews of market-based instruments mention several US environmental markets, like the Acid Rain Program (Stavins 2003). Our analysis of offset markets represents a far larger number of distinct environmental markets than has been previously studied. The granularity and spatial diversity of these markets is also important for our study of efficiency, as regulatory stringency also differs by location, time, and pollutant.

Appendix Figure 2 maps locations of these markets. These maps show counties in states with policies that set up offset markets and that have been part of a nonattainment area at any time in 2010–2019. Panel A of Appendix Figure 2 describes markets for NO_x and VOCs, corresponding to nonattainment for ground-level ozone. More markets have existed for these pollutants than for others. Panel B shows areas with markets for other pollutants; the most common is for particulate matter, but some offset markets also cover carbon monoxide and sulfur oxides. Supplemental Appendix C.2 describes other measures of market size and importance.

B. Trends in Offset Prices

To understand the evolution of regulatory stringency over the past 30 years, we explore trends in mean offset prices:

$$O_{mpy} = \beta y_y + \mu_{mp} + \epsilon_{mpy}.$$

Here O represents the ton-weighted mean log offset price for market m , pollutant p , and year y . We regress this on a linear year trend y_y and market $\times$ pollutant fixed effects μ_{mp} , with error ϵ_{mpy} . The coefficient β represents the mean annual change in log offset prices within a market $\times$ pollutant.

Appendix Table 2 presents estimates of β . Panel A pools pollutants and markets. Mean real offset prices increase by 5–8 percent per year. The pattern is similar whether we treat each offset transaction as having equal weight (column 1) or weight by tons or population (columns 2 and 3). In panels B and C, which show each pollutant separately, the two pollutants have similar trends. Appendix Table 2 includes all years with data; limiting the sample to the years 2010–2019 delivers smaller annual growth rates. The growth rate in offset prices is similar to recent rates of return in equity markets (Jordà et al. 2019). Offsets, like stocks, have short- and long-run price volatility reflecting shifts in demand or supply.

C. Relationship between Offset Prices and Measures of Environmental Compliance

We now describe relationships between offset prices and commonly-studied measures of regulatory stringency or compliance, such as county-level nonattainment severity, abatement expenditures, emission levels, and output of polluting industries (Greenstone 2002; Becker 2005; Shapiro and Walker 2018). Supplemental Appendix E.1 discusses methods.

We first compare offset prices to rates of industrial spending on pollution abatement as reported by administrative versions of the US Census Bureau's Pollution Abatement Control and Expenditures Survey. Appendix Table 1 shows large and statistically significant positive relationships between offset prices and abatement expenditures. Row 1, column 1 implies that for plants that emit NO_x , a 1 percent increase in abatement operating expenditures per dollar of output is associated with a 1.8 percent increase in the NO_x offset price. Row 2 finds a larger elasticity for pollution abatement operating expenditures divided by value added. Magnitudes differ by pollutant and independent variable.

We also compare offset prices to emissions intensity using data from the EPA's National Emissions Inventory on the tons of pollution emitted per dollar of output or dollar of value added (US Environmental Protection Agency 2020a, b, c). Appendix Table 2 suggests that regions with lower emissions intensity have higher offset prices. For example, column 1 of panel A implies that a 10 percent increase in the emissions intensity of NO_x -emitting establishments is associated with a 1.69 percent reduction in the NO_x offset price. These elasticities are similar across pollutants and when looking at emissions per unit of output or emissions per unit of value added, with an elasticity around -0.14 .

Discussions with market participants suggests that growing demand for offsets siphons off cheap abatement opportunities and low cost offsets. Accordingly, offset prices rise as remaining abatement opportunities become more costly. Offset market participants in our interviews often noted this phenomenon. For example, lower natural gas prices due to hydraulic fracturing (fracking) increased demand for petrochemicals production in Houston, raising offset prices. Appendix Table 3, panel A, finds that offset prices are associated with increases in local value added from polluters. Column 1 shows that a 10 percent increase in value added from NO_x -emitting facilities is associated with a 13 percent increase in mean NO_x offset prices within an air district or an elasticity of 1.3. Column 2 finds a slightly larger though imprecisely estimated elasticity for VOCs. Column 3 pools pollutants and shows that the average long-run elasticity is around 1.4. In panel B the relationships

between offset prices and output are weaker and statistically insignificant compared to estimates using industry value added.

Finally, we compare offset prices against nonattainment stringency. The 1990 Clean Air Act Amendments introduced a new ozone standard including classifications for ozone nonattainment severity. Depending on ambient ozone levels, areas could be classified into five nonattainment classifications: extreme, severe, serious, moderate, and marginal. Increasing nonattainment stringency can affect offset markets for ozone precursors, NO_x and VOC, for two reasons. First, under more stringent nonattainment classifications, the EPA broadens the set of polluting facilities that must purchase offsets. Second, as nonattainment severity increases, facilities become required to purchase proportionally *more* offsets than they emit depending on severity-specific trading ratios. These trading ratios range from 1-to-1 for marginal nonattainment to 1.5-to-1 for extreme nonattainment. Supplemental Appendix Table 4 finds that more stringent nonattainment classifications have higher offset prices. Column 1 shows that counties in moderate nonattainment have 16 percent higher offset prices than counties in marginal nonattainment though this difference is not statistically significant at conventional levels. Offset prices increase monotonically with nonattainment stringency. Offset transactions in extreme nonattainment areas sell for nearly five times higher prices per ton than offsets in marginal nonattainment areas.¹⁰ Columns 2–4 obtain similar results from adding fixed effects for years, pollutants, or pollutant $\times$ year combinations.

IV. Efficiency: Comparing Offset Prices to Marginal Benefits of Abatement

Appendix Table 3 compares the marginal benefits of pollution abatement to offset prices, pooling data from 16 states plus Washington, DC, over the years 2010–2019. Columns 1 and 2 describe NO_x ; columns 3 and 4 describe VOCs. Columns 1 and 3 weight across markets by tons of pollution, and columns 2 and 4 weight by population. Panel A pools markets. Panels B–E describe each US census region. Within a panel, row 1 shows mean marginal benefits of abatement divided by mean offset prices, row 2 shows the p -value for the hypothesis test that this ratio equals one, row 3 describes mean marginal benefits of abatement, and row 4 describes mean offset prices. Under the interpretation that mean offset prices equal marginal abatement costs, row 1 represents the ratio of marginal benefits to marginal costs of pollution abatement. Supplemental Appendix Tables 5 and 6 present estimates using the maximum offset price and median offset price within a district, respectively.

Appendix Table 3 shows that national mean marginal benefits of abatement substantially exceed offset prices. This provides our main result that air pollution regulation is typically less stringent than economic efficiency would require. This conclusion is statistically significant at greater than 99 percent confidence for most pollutants and regions. For NO_x , mean marginal benefits of abatement are \$42,000–\$50,000, depending on whether they are weighted by pollution or population. Mean offset prices are \$1,100–\$4,300. Thus, the ratio of marginal benefits of abatement to offset prices is 12 to 37. For VOCs, we obtain a ratio of 9 to 29.

¹⁰We calculate this from $\exp(1.725) - 1 \approx 4.6$.

To better interpret these ratios, consider an incumbent firm deciding whether to permanently decrease NO_x emissions and generate offsets to sell. On average, the firm would receive \$1,100–\$4,300 per ton for decreasing pollution. However, these same emissions reductions would create \$42,000–\$50,000 per ton in health benefits to society. In this sense, regulation gives less incentive to clean up pollution than is optimal, so it is too lenient.

Panels B–E of Appendix Table 3 show similar patterns in all four census regions. For NO_x and VOCs, both weighting schemes, and all four regions, the ratio of the marginal benefits of abatement to offset prices far exceeds one. For NO_x in the northeast, for example, the marginal benefits of abatement are \$44,000 but mean offset prices are \$600. Ratios are modestly lower in the west, ranging from 4.6 to 8.2. Ratios are the lowest in the south, ranging from 2.3 to 20.4.

Appendix Figure 1 plots mean offset prices and the marginal benefits of abatement for all markets (panel A) and each census region (panels B–E), by year. The marginal benefits of abatement vary year by year due to changes in population and differences in baseline levels of all pollutants. Appendix Table 3 essentially shows the mean value of these lines in the 2010–2019 period, while these graphs show the underlying year-by-year averages for all years.

Appendix Figure 1 reveals the large gaps between the marginal benefits of abatement and offset prices in most regions, pollutants, and years. That gap reflects the finding that the marginal benefits of abatement dramatically exceed mean offset prices. The only exception is for VOCs in the south, where the marginal benefits of abatement and offset prices are closer. The yearly values in Appendix Figure 1 are near decadal means from Appendix Table 3.

Supplemental Appendix Table 7 shows the ratio of the marginal benefits of abatement to offset prices for each of our largest markets, separately for NO_x and VOCs. These markets are heterogeneous—they include longstanding industrial cities like Cleveland and Pittsburgh; faster-growing, high-education cities, like Los Angeles and Washington, DC; and more rural areas like the Central Valley of California and the Upper Green River Basin in Wyoming.

Given this heterogeneity, it is striking that many markets have high ratios of the marginal benefits of abatement to offset prices. Across markets in Supplemental Appendix Table 7, the median ratio is 40. In about three-fourths of the markets, this ratio exceeds 10. The only markets with a ratio below four are a few in California, the markets in Houston, and one market in Wyoming. Even for these markets with lower ratios, only one of the forty markets listed in Supplemental Appendix Table 7 has a ratio below one—VOCs in Houston.

What drives these differences? Appendix Figure 2 suggests that differences across markets are more driven by variation in the marginal benefits of abatement than by offset prices. This is because the marginal benefits of abatement (hollow red diamonds in the graph) vary widely between markets, from \$1,000/ton for VOCs in Wyoming to \$100,000/ton for NO_x in Los Angeles. By contrast, offset prices (the solid blue circles in the graph) vary less in dollar terms, though more in percentages, from \$100 in many markets to \$10,000 for NO_x in Los Angeles. Appendix Figure 2 also shows that for Houston VOCs, high offset prices rather than low marginal abatement benefits make the marginal benefits of abatement less than offset prices.

Appendix Figure 3 graphs year-by-year trends in the marginal benefits of abatement and the distribution of offset prices in the six largest markets where we observe transaction-level offset prices. Panel A presents these graphs, pooling markets. Subsequent panels break out these trends by market. Blue bars show the tenth to ninetieth percentile range of offset prices within a market $\times$ pollutant $\times$ year, thick black horizontal lines within each blue bar show mean offset prices in each market $\times$ pollutant $\times$ year, and red dashed lines show marginal benefits of abatement. Panel A shows that for most years the ninetieth percentile offset price is below the marginal benefit of abatement. Starting in 2014, the upper end of the VOC price distribution exceeds the average marginal benefit of abatement. This mostly comes from the Houston VOCs market, as shown in panel E; prices in Houston were below \$1,000 until 2010 then skyrocketed to over \$10,000 per ton.

A few forces explain why there is price variation within a market $\times$ year. First, these markets could have search frictions. Supplemental Appendix B.5 discusses offset price dispersion in the context of search frictions. Price dispersion may also emerge from heterogeneity in marginal abatement costs; as more abatement opportunities are exhausted within the marketplace, it becomes increasingly costly to abate. Similarly, intra-annual variation in supply and demand can generate equilibrium price volatility throughout the year.

Houston VOC prices are over six times the price of VOC offsets elsewhere. Houston's offset prices for NO_x are also high. This may seem surprising since Texas traditionally has weaker political support for environmental regulation than other states. Several reasons explain this apparent anomaly. First, growing demand for petrochemical, energy, and related industries to enter the Houston market substantially increased offset prices after 2010. Second, because the Clean Air Act is federal, states have limited discretion over precise standards regulating each plant. If states or localities do not regulate up to the federal standard, they face strict sanctions and threats of federal takeover of state regulatory implementation.

Supplemental Appendix E.2 investigates sensitivity to alternative ways of summarizing offset prices. These alternative estimates generally change the ratio of offset prices to marginal abatement benefits in intuitive ways. Despite varying magnitudes, most qualitative patterns persist. Nearly all ratios far exceed one; the main exception is the Houston VOC market. Supplemental Appendix E.2 also explores sensitivity to alternative estimates of the marginal benefits of pollution abatement. Most conclusions are qualitatively unchanged.

We use bounding to ask what VSL implies that regulation is efficient, that is, that the ratio of marginal benefits to offset prices equals one. For NO_x , the implied VSL is \$0.5 million. For VOCs, it is \$1.1 million. These VSL bounds are considerably lower than the EPA's \$8.8 million VSL.

V. Conclusion

US air pollution offset markets are a key part of the Clean Air Act but have received little attention from academics. We analyze offsets and describe how they can help assess the efficiency of air pollution regulation using decentralized transactions data from many markets. This approach has appealing features: It

uses revealed preference; reflects pecuniary and nonpecuniary abatement costs; obtains estimates that vary by pollutant, location, and time; and is straightforward to implement.

We find that the marginal benefit of pollution abatement far exceeds mean offset prices as indicated by a wide range of air quality and valuation models. Using the intuition that prices in environmental markets reveal marginal abatement costs, this leads to our main policy conclusion that policy is more lenient than is efficient in most markets. Regulation in the Houston market for VOCs, however, may be moderately more stringent than efficiency would require. Outside Houston, unless marginal abatement costs rise or marginal benefits fall substantially (neither of which seems likely in plausible future US policy), Clean Air Act regulations are far from efficient. This general conclusion is in line with benefit-cost analyses. While these analyses generally rely on engineering estimates, they often conclude that incremental Clean Air Act regulations have a benefit/cost ratio above ten (Keiser and Shapiro 2019).

Our conclusions about the stringency of environmental policy are not made in a vacuum. Reforms reducing compliance costs of the Clean Air Act or larger estimates of the marginal benefits of abatement would strengthen our conclusions. Alternatively, if marginal abatement costs continue increasing at their historical rate, marginal abatement costs could begin exceeding the benefits to society, and regulation would become more stringent than is efficient. At the same time, as firms face increasing compliance costs, the incentive for finding creative solutions to pollution abatement increases. Innovations in abatement technology would change the efficient level of environmental stringency.

We believe this paper offers several questions for future work. How can analysis of offset markets for other environmental goods or outside the United States shed light on the efficiency of regulation? What other revealed preference strategies can estimate the marginal costs of pollution abatement? Such strategies could help compare abatement costs under different market designs and in areas without offset markets. Research has made extraordinary progress in measuring the marginal benefits of pollution abatement but has made less progress in understanding the marginal costs of pollution abatement, though both parameters are essential to designing optimal policy.

APPENDIX A. FIGURES AND TABLES

TABLE 1—PREVALENCE OF OFFSET MARKETS

	Number of markets	Population (millions)		GDP (trillions)		Manufacturing employment (millions)	
		People	Percent	Dollars	Percent	Workers	Percent
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A. National</i>							
Any pollutant	491	182.1	59	11.09	66	6.44	56
Ozone	282	173.2	56	10.66	63	6.09	53
Particulate matter	83	121.2	39	7.59	45	3.97	34
<i>Panel B. National—analysis period</i>							
Any pollutant	226	158.1	51	9.75	58	5.38	47
Ozone	118	145.0	47	9.09	54	4.83	42
Particulate matter	63	114.3	37	7.12	42	3.74	32
<i>Panel C. Full sample (16 states plus Washington, DC) as proportion of all national markets</i>							
Any pollutant	42	94.5	60	6.15	63	3.00	56
Ozone	37	94.5	65	6.15	68	3.00	62
Particulate matter	5	30.3	27	1.84	26	1.05	28

Notes: This table describes all US air pollution offset markets. Percentages in panel C describe the sample as a share of all national offset markets. A market is a distinct nonattainment area $\times$ pollutant in states with offset markets, designated for nonattainment in any part of years 1992–2019 (panel A) or 2010–2019 (panel B). Ozone nonattainment areas have separate markets for nitrogen oxides and volatile organic compounds. Nitrogen dioxide markets are included in ozone. “Any pollutant” includes carbon monoxide, lead, and sulfur dioxide, in addition to ozone. Population, GDP, and employment represent the year 2010 and include any county which has a market for at least one pollutant. Population data are from the population census, county GDP data are from Bureau of Economic Analysis Regional Economic Accounts (Bureau of Economic Analysis 2020), and manufacturing employment data are from the US Bureau of Labor Statistics Quarterly Census of Employment and Wages (US Bureau of Labor Statistics 2020b). GDP is deflated to 2017 dollars using the GDP deflator.

TABLE 2—TRENDS IN OFFSET PRICES

	Offset prices		
	(1)	(2)	(3)
<i>Panel A. All pollutants</i>			
Year	0.05 (0.02)	0.08 (0.06)	0.06 (0.02)
Observations	208	208	208
<i>Panel B. Nitrogen oxides (NO_x)</i>			
Year	0.05 (0.02)	0.08 (0.02)	0.06 (0.02)
Observations	98	98	98
<i>Panel C. Volatile organic compounds (VOCs)</i>			
Year	0.05 (0.03)	0.08 (0.09)	0.05 (0.03)
Observations	110	110	110
Weight		Tons	Population

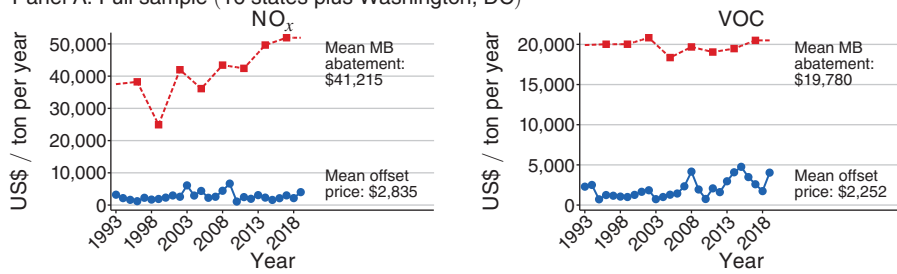
Notes: Each unit of observation is a market $\times$ pollutant $\times$ year. Dependent variables in logs. For observations at the nonattainment area $\times$ pollutant $\times$ year level, offset prices are either mean weighted by tons traded or by county population, as indicated by the last row. All estimates include market $\times$ pollutant fixed effects. Standard errors are clustered within each market $\times$ pollutant.

TABLE 3—RATIO OF MARGINAL BENEFITS OF ABATEMENT TO MEAN OFFSET PRICES, 2010–2019

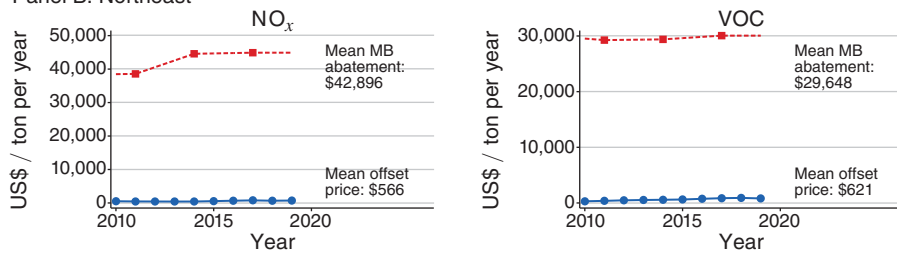
	NO _x		VOCs	
	(1)	(2)	(3)	(4)
<i>Panel A. Full sample (16 states plus Washington, DC)</i>				
1. Marginal benefits of abatement/Offset price	37.31	11.63	29.04	8.96
2. <i>p</i> -value: MBabatement/Offset price = 1	[0.00]	[0.00]	[0.00]	[0.00]
3. Mean marginal benefits of abatement	\$42,035	\$50,434	\$24,807	\$20,593
4. Mean offset prices	\$1,127	\$4,335	\$854	\$2,298
<i>Panel B. Northeast</i>				
1. Marginal benefits of abatement/Offset price	77.29	68.66	46.63	51.77
2. <i>p</i> -value: MBabatement/Offset price = 1	[0.00]	[0.00]	[0.00]	[0.00]
3. Mean marginal benefits of abatement	\$43,784	\$42,896	\$28,980	\$31,953
4. Mean offset prices	\$566	\$625	\$621	\$617
<i>Panel C. South</i>				
1. Marginal benefits of abatement/Offset price	20.35	5.87	11.66	2.29
2. <i>p</i> -value: MBabatement/Offset price = 1	[0.00]	[0.01]	[0.02]	[0.23]
3. Mean marginal benefits of abatement	\$37,225	\$28,684	\$19,919	\$13,431
4. Mean offset prices	\$1,829	\$4,886	\$1,709	\$5,856
<i>Panel D. West</i>				
1. Marginal benefits of abatement/Offset price	8.19	8.14	4.64	6.62
2. <i>p</i> -value: MBabatement/Offset price = 1	[0.00]	[0.00]	[0.00]	[0.00]
3. Mean marginal benefits of abatement	\$35,256	\$70,081	\$7,433	\$15,596
4. Mean offset prices	\$4,304	\$8,610	\$1,601	\$2,356
<i>Panel E. Midwest</i>				
1. Marginal benefits of abatement/Offset price	84.43	89.14	45.98	49.00
2. <i>p</i> -value: MBabatement/Offset price = 1	[0.04]	[0.02]	[0.02]	[0.01]
3. Mean marginal benefits of abatement	\$44,122	\$48,114	\$19,841	\$22,302
4. Mean offset prices	\$523	\$540	\$432	\$455
<i>Weight:</i>				
Tons	X		X	
Population		X		X

Notes: Offset prices and marginal benefits of abatement (MBabatement) are in dollars per ton of emissions. Row 1 in each panel shows the ratio of marginal benefits of abating one ton of emissions to mean offset prices per ton of emissions. Row 2 shows the *p*-value for testing the null hypothesis that the ratio in row 1 equals one. Rows 3 and 4 show the mean marginal benefits of abatement and mean offset prices, respectively. Data represent the years 2010–2019. Offset prices are the mean price of pollution offsets per ton for the indicated census region, pollutant, and time period, weighted by transaction amount in tons or by population in offset markets and annualized using the price ratio between permanent and temporary offset prices. Data on marginal benefits are available for the years 2008, 2011, 2014, and 2017 and linearly interpolated between years. All currency are in 2017 dollars, deflated using the GDP deflator. Abatement marginal benefits for each region are weighted across counties within a market according to county population in the 2010 census (US Census Bureau 2010) and weighted across markets by transaction amount in tons or by population in offset markets.

Panel A. Full sample (16 states plus Washington, DC)



Panel B. Northeast



Panel C. South

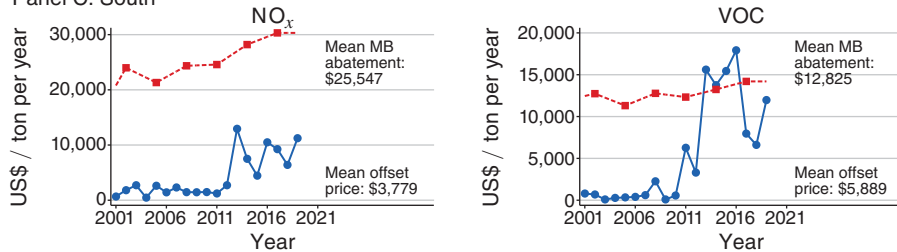
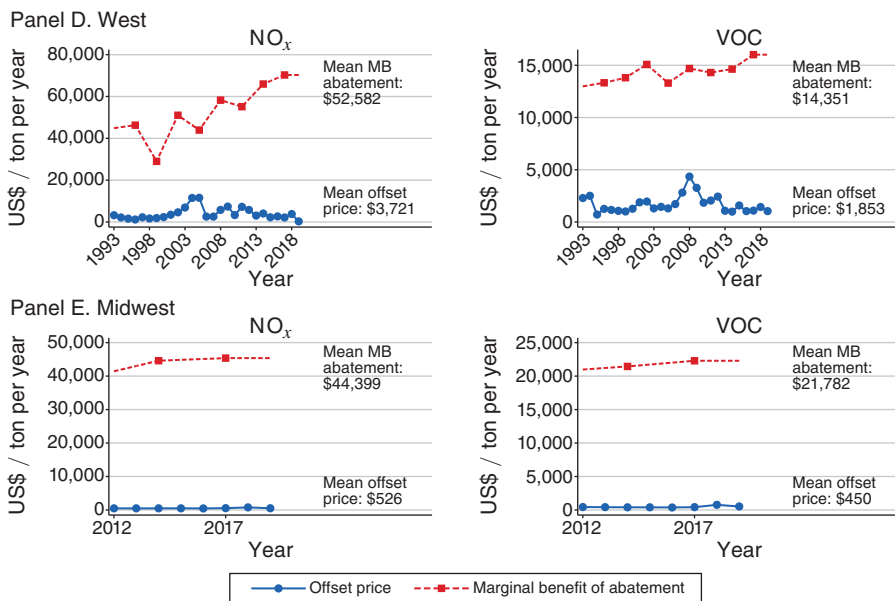


FIGURE 1. POLLUTION OFFSET PRICES VERSUS MARGINAL BENEFITS OF ABATEMENT BY YEAR

(continues)

FIGURE 1. POLLUTION OFFSET PRICES VERSUS MARGINAL BENEFITS OF ABATEMENT BY YEAR (*continued*)

Notes: These graphs show pollution offset prices and the marginal benefits of pollution abatement by year with separate graphs for each pollutant and census region. Blue solid lines show mean offset prices in each market $\times$ pollutant $\times$ year; red dashed lines show marginal benefits of abatement. Offset prices are the mean price of pollution offsets per ton for the indicated nonattainment area, pollutant, and time period, weighted by transaction amount in short tons and annualized using the observed price ratio between permanent and temporary offsets. Marginal benefits of abatement are the marginal external costs avoided per short ton abated for the indicated nonattainment area and pollutant for the years 2008, 2011, 2014, and 2017 and linearly interpolated between years. Marginal benefits of abatement are weighted across counties within an offset market according to county population in the 2010 census (US Census Bureau 2010). All monetary amounts are in 2017 dollars deflated by the Federal Reserve's US GDP deflator.

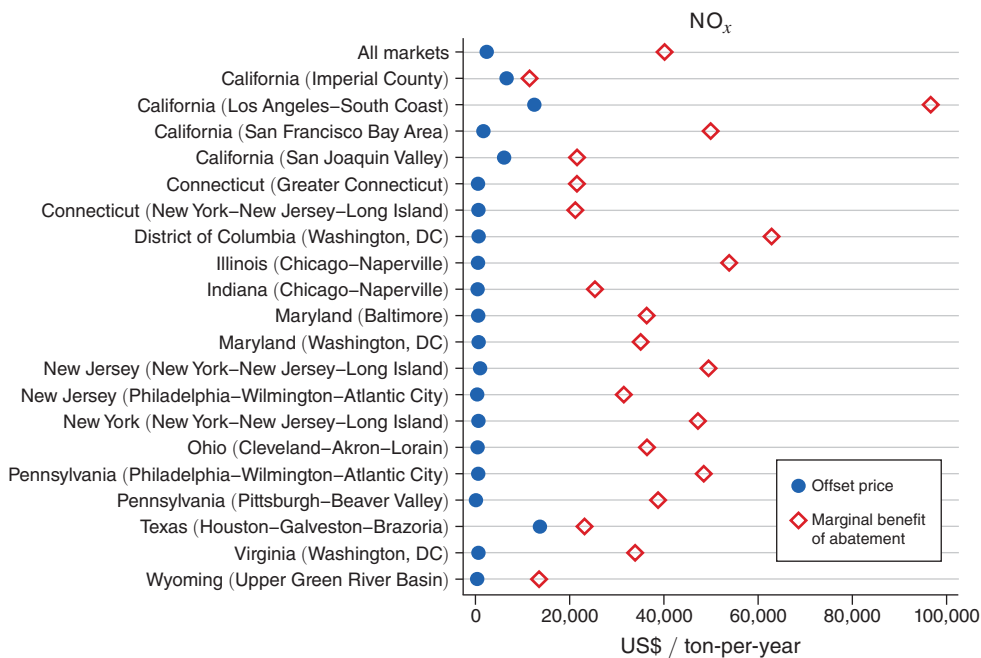


FIGURE 2. OFFSET PRICES AND MARGINAL BENEFITS OF ABATEMENT, LARGE INDIVIDUAL MARKETS

(continues)

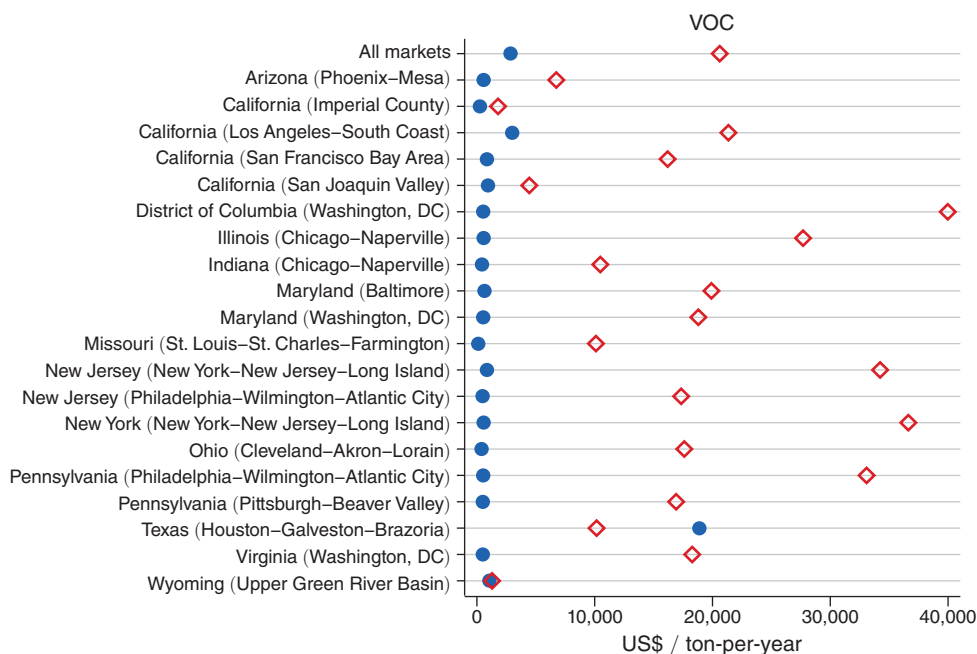


FIGURE 2. OFFSET PRICES AND MARGINAL BENEFITS OF ABATEMENT, LARGE INDIVIDUAL MARKETS (*continued*)

Notes: This figure compares offset prices and the marginal benefits of pollution abatement in individual market $\times$ pollutants for a set of large markets with data. Data represent the years 2010–2019. The vertical axis lists the states that the data represent with the market's name in parentheses. Offset prices are the mean price of pollution offsets per ton for the indicated census region, pollutant, and time period, weighted by transaction amount in tons or by population in offset markets and annualized using the price ratio between permanent and temporary offset prices. Marginal benefits of abatement are the marginal external costs avoided per ton abated for the indicated market and pollutant. Data on marginal benefits are available for the years 2008, 2011, 2014, and 2017 and linearly interpolated between years. All monetary amounts are in 2017 dollars deflated using the GDP deflator. Abatement marginal benefits for each market are weighted across counties within a market according to county population in the 2010 census (US Census Bureau 2010). The Philadelphia–Wilmington–Atlantic City area includes Delaware.

REFERENCES

- AEA Technology. 2001. *The Costs of Reducing PM10 and NO2 Emissions and Concentrations in the UK: Part 1: PM10*. AEA Technology.
- Aldy, Joseph, Matthew Kotchen, Mary Evans, Meredith Fowlie, Arik Levinson, and Karen Palmer. 2020. "Deep Flaws in a Mercury Regulatory Analysis." *Science* 368 (6488): 247–48.
- Allcott, Hunt, and Michael Greenstone. 2023. "Measuring the Welfare Effects of Residential Energy Efficiency Programs." NBER Working Paper 23386.
- Arias, Elizabeth. 2014. "United States Life Tables, 2010." *National Vital Statistics Reports* 63 (7): 1–63.
- Bajari, Patrick, Jane Cooley Fruehwirth, Kyoo il Kim, and Christopher Timmins. 2012. "A Rational Expectations Approach to Hedonic Price Regression with Time-Varying Unobserved Product Attributes: The Price of Pollution." *American Economic Review* 102 (5): 1898–926.
- Becker, Randy A. 2005. "Air Pollution Abatement Costs under the Clean Air Act: Evidence from the Pace Survey." *Journal of Environmental Economics and Management* 50 (1): 144–69.
- Berman, Eli, and Linda T. M. Bui. 2001. "Environmental Regulation and Productivity: Evidence from Oil Refineries." *Review of Economics and Statistics* 83 (3): 498–510.
- Bureau of Economic Analysis. 2020. *Regional Economic Accounts*. <https://apps.bea.gov/regional/downloadzip.cfm> (accessed January 26, 2020).

- California Air Resources Board.** 2019. *New Source Review - Emission Reduction Credit Offsets*. <https://ww2.arb.ca.gov/new-source-review-emission-reduction-credit-offsets> (accessed June 25, 2019).
- California Air Resources Board.** 2020. *Facility Search Engine*. <https://ww2.arb.ca.gov/applications/facility-search-engine> (accessed January 26, 2020).
- Carleton, Tamma, Amir Jina, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, et al.** 2022. "Valuing the Global Mortality Consequences of Climate Change." *Quarterly Journal of Economics* 137 (4): 2037–105.
- Carlson, Curtis, Dallas Burtraw, Maureen Cropper, and Karen L. Palmer.** 2000. "Sulfur Dioxide Control by Electric Utilities: What Are the Gains from Trade?" *Journal of Political Economy* 108 (6): 1292–326.
- Center for Air, Climate and Energy Solutions.** 2020. *CACES RCM/LUR Data Download*. <https://www.caces.us/data> (accessed January 26, 2020).
- Centers for Disease Control and Prevention.** 2020a. *Multiple Cause of Death, 2007–2020 Request*. CDC WONDER. <https://wonder.cdc.gov/mcd-icd10.html> (accessed June 29, 2020).
- Centers for Disease Control and Prevention.** 2020b. *Natality, 2007–2020 Request*. CDC WONDER. <https://wonder.cdc.gov/controller/datarequest/D66> (accessed March 20, 2020).
- Chay, Kenneth Y., and Michael Greenstone.** 2005. "Does Air Quality Matter? Evidence from the Housing Market." *Journal of Political Economy* 113 (2): 376–424.
- Crocker, Thomas D.** 1966. "The Structuring of Atmospheric Pollution Control Systems." In *The Economics of Air Pollution*, edited by H. Wlozin, 61–86. W. W. Norton & Co.
- Cropper, Maureen L., and Wallace E. Oates.** 1992. "Environmental Economics: A Survey." *Journal of Economic Literature* 30 (2): 675–740.
- Currie, Janet, Lucas Davis, Michael Greenstone, and Reed Walker.** 2015. "Environmental Health Risks and Housing Values: Evidence from 1,600 Toxic Plant Openings and Closings." *American Economic Review* 105 (2): 678–709.
- Currie, Janet, and Reed Walker.** 2019. "What Do Economists Have to Say about the Clean Air Act 50 Years after the Establishment of the Environmental Protection Agency?" *Journal of Economic Perspectives* 33 (4): 3–26.
- Dales, J. H.** 1968. *Pollution, Property, and Prices: An Essay in Policy-Making and Economics*. University of Toronto Press.
- Deschênes, Olivier, Michael Greenstone, and Joseph S. Shapiro.** 2017. "Defensive Investments and the Demand for Air Quality: Evidence from the NO_x Budget Program." *American Economic Review* 107 (10): 2958–89.
- Dominici, Francesca, Michael Greenstone, and Cass R. Sunstein.** 2014. "Particulate Matter Matters." *Science* 344 (6181): 257–59.
- Dudek, Daniel J., and John Palmisano.** 1988. "Emissions Trading: Why Is This Thoroughbred Hobbled?" *Columbia Journal of Environmental Law* 13 (2): 217–56.
- DuPuis, E. Melanie.** 2004. "Who Owns the Air? Clean Air Act Implementation as a Negotiation of Common Property Rights." In *Smoke and Mirrors: The Politics and Culture of Air Pollution*, edited by E. Melanie DuPuis, 223–40. NYU Press.
- Dwyer, John P.** 1992. "California's Tradable Emissions Policy and Greenhouse Gas Control." *Journal of Energy Engineering* 118 (2): 59–76.
- Ebenstein, Avraham, Maoyong Fan, Michael Greenstone, Guojun He, and Maigeng Zhou.** 2017. "New Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *Proceedings of the National Academy of Sciences* 114 (39): 10384–89.
- ECR Incorporated.** 1998. *Stational Source Control Techniques Document for Fine Particulate Matter*. US Environmental Protection Agency.
- Ellerman, A. Danny, Paul L. Joskow, and David Harrison Jr.** 2003. *Emissions Trading in the US: Experience, Lessons, and Considerations for Greenhouse Gases*. Pew Center on Global Climate Change.
- Emission Advisors Inc.** 2020. *ERC Pricing History 2010–2019*.
- Fann, Neal, Amy D. Lamson, Susan C. Anenberg, Karen Wesson, David Risley, and Bryan J. Hubbell.** 2012. "Estimating the National Public Health Burden Associated with Exposure to Ambient PM_{2.5} and Ozone." *Risk Analysis* 32 (1): 81–95.
- Federal Reserve Bank of St. Louis.** 2020a. *Gross Domestic Product: Implicit Price Deflator (GDP-DEF)*. Federal Reserve Economic Data. <https://fred.stlouisfed.org/series/GDPDEF#0> (accessed April 17, 2020).
- Federal Reserve Bank of St. Louis.** 2020b. *Producer Price Index by Industry: Total Manufacturing Industries (PCUOMFGOMFG)*. Federal Reserve Economic Data. <https://fred.stlouisfed.org/series/PCUOMFGOMFG#0> (accessed January 16, 2020).

- Fort, Jeffrey C., and Cynthia A. Faur.** 1997. "Can Emissions Trading Work beyond a National Program?: Some Practical Observations on the Available Tools." *University of Pennsylvania Journal of International Law* 18 (2): 463–76.
- Foster, Vivien, and Robert W. Hahn.** 1995. "Designing More Efficient Markets: Lessons from Los Angeles Smog Control." *Journal of Law and Economics* 38 (1): 19–48.
- Fowlie, Meredith, Michael Greenstone, and Catherine Wolfram.** 2018. "Do Energy Efficiency Investments Deliver? Evidence from the Weatherization Assistance Program." *Quarterly Journal of Economics* 133 (3): 1597–644.
- Fowlie, Meredith, Christopher R. Knittel, and Catherine Wolfram.** 2012. "Sacred Cars? Cost-Effective Regulation of Stationary and Nonstationary Pollution Sources." *American Economic Journal: Economic Policy* 4 (1): 98–126.
- Fraas, Art, John D. Graham, and Jeff Holmstead.** 2017. "EPA's New Source Review Program: Time for Reform?" *Environmental Law Reporter* 47 (1): 10026.
- Fromm, O., and B. Hansjürgens.** 1996. "Emission Trading in Theory and Practice: An Analysis of RECLAIM in Southern California." *Environment and Planning C: Politics and Space* 14 (3): 367–84.
- Gauna, Eileen.** 1996. "Major Sources of Criteria Pollutants in Nonattainment Areas: Balancing the Goals of Clean Air, Environmental Justice, and Industrial Development." *Hastings West-Northwest Journal of Environmental Law and Policy* 3 (3): 379–404.
- General Accounting Office.** 1982. *A Market Approach to Air Pollution Control Could Reduce Compliance Costs without Jeopardizing Clean Air Goals*. GAO.
- Giglio, Stefano, Matteo Maggiori, Krishna Rao, Johannes Stroebe, and Andreas Weber.** 2021. "Climate Change and Long-Run Discount Rates: Evidence from Real Estate." *Review of Financial Studies* 34 (8): 3527–71.
- Gilmore, Elisabeth A., Jinhyok Heo, Nicholas Z. Muller, Christopher W. Tessum, Jason D. Hill, Julian D. Marshall, and Peter J. Adams.** 2019. "An Inter-Comparison of the Social Costs of Air Quality from Reduced-Complexity Models." *Environmental Research Letters* 14 (7): 74016.
- Gollop, Frank M., and Mark J. Roberts.** 1985. "Cost-Minimizing Regulation of Sulfur Emissions: Regional Gains in Electric Power." *Review of Economics and Statistics* 67 (1): 81–90.
- Greenstone, Michael.** 2002. "The Impacts of Environmental Regulations on Industrial Activity: Evidence from the 1970 and 1977 Clean Air Act Amendments and the Census of Manufactures." *Journal of Political Economy* 110 (6): 1175–219.
- Haddad, Brent M., and John Palmisano.** 2001. "Market Darwinism versus Market Creationism: Adaptability and Fairness in the Design of Greenhouse Gas Trading Mechanisms." *International Environmental Agreements: Politics, Law and Economics* 1 (4): 427–46.
- Hahn, Robert W., and Gordon L. Hester.** 1987. "The Market for Bads: EPA's Experience with Emissions Trading." *Regulation* 11 (3–4): 48–53.
- Hahn, Robert W., and Gordon L. Hester.** 1989. "Where Did All the Markets Go—An Analysis of EPA's Emissions Trading Program." *Yale Journal on Regulation* 6 (1): 109–54.
- Heo, Jinhyok, Peter J. Adams, and H. Oliver Gao.** 2016. "Reduced-Form Modeling of Public Health Impacts of Inorganic PM_{2.5} and Precursor Emissions." *Atmospheric Environment* 137: 80–89.
- Holland, Stephen P., Erin T. Mansur, Nicholas Z. Muller, and Andrew J. Yates.** 2019. "Distributional Effects of Air Pollution from Electric Vehicle Adoption." *Journal of the Association of Environmental and Resource Economists* 6 (S1): S65–S94.
- Holland, Stephen P., Erin T. Mansur, Nicholas Z. Muller, and Andrew J. Yates.** 2020. "Decompositions and Policy Consequences of an Extraordinary Decline in Air Pollution from Electricity Generation." *American Economic Journal: Economic Policy* 12 (4): 244–74.
- Huang, Ye, Huizhong Shen, Han Chen, Rong Wang, Yanyan Zhang, Shu Su, Yuanchen Chen, et al.** 2014. "Quantification of Global Primary Emissions of PM_{2.5}, PM₁₀, and TSP from Combustion and Industrial Process Sources." *Environmental Science and Technology* 48 (23): 13834–43.
- Jaeger, David A., and Karl Storchmann.** 2011. "Wine Retail Price Dispersion in the United States: Searching for Expensive Wines?" *American Economic Review* 101 (3): 136–41.
- Jordà, Óscar, Katharina Knoll, Dmitry Kuvshinov, Moritz Schularick, and Alan M. Taylor.** 2019. "The Rate of Return on Everything, 1870–2015." *Quarterly Journal of Economics* 134 (3): 1225–98.
- Keiser, David A., and Joseph S. Shapiro.** 2019. "US Water Pollution Regulation over the Last Half Century: Burning Waters to Crystal Springs?" *Journal of Economic Perspectives* 33 (4): 51–75.
- Klimont, Zbigniew, Janusz Cofala, Imrich Bertok, Markus Amann, Chris Heyes, and Frantisek Gyarfas.** 2002. *Modelling Particulate Emissions in Europe: A Framework to Estimate Reduction Potential and Control Costs*. International Institute for Applied Systems Analysis.
- Kolstad, Charles D., and Michael D. Williams.** 1989. "Aggregate Source-Receptor Relations for Economic Analysis of Ambient Regulations." *Journal of the Air and Waste Management Association* 39 (6): 824–30.

- Krewski, D., R. T. Burnett, M. S. Goldberg, K. Hoover, J. Siemiatycki, M. Jerrett, M. Abrahamowicz, W. H. White, G. Bartlett, and L. Brodsky. 2009. *Extended Follow-Up and Spatial Analysis of the American Cancer Society Study Linking Particulate Air Pollution and Mortality*. Health Effects Institute.
- Lange, Sabine S., Sean E. Mulholland, and Michael E. Honeycutt. 2018. "What Are the Net Benefits of Reducing the Ozone Standard to 65 ppb? An Alternative Analysis." *International Journal of Environmental Research and Public Health* 15 (8): 1586.
- Leonard, R. Leon. 1996. *Air Quality Permitting*. CRC Press.
- Leonard, R. Leon. 2018. *Air Quality Permitting*. Routledge. VitalSource eBooks.
- Lepeule, Johanna, Francine Laden, Douglas Dockery, and Joel Schwartz. 2012. "Chronic Exposure to Fine Particles and Mortality: An Extended Follow-Up of the Harvard Six Cities Study from 1974 to 2009." *Environmental Health Perspectives* 120: 965–70.
- Livemore, Michael A., and Richard L. Revesz. 2015. "EPA's Ozone Standard Is Insufficiently Stringent, Not Overly Expensive." *Regulatory Review*, February 16. <https://www.theregreview.org/2015/02/16/livemore-revesz-epa-ozone-standard/>.
- Montgomery, W. David. 1972. "Markets in Licenses and Efficient Pollution Control Programs." *Journal of Economic Theory* 5 (3): 395–418.
- Morss, Elizabeth, and David R. Wooley. 2022. *Clean Air Act Handbook*. 32nd ed. Thomson Reuters.
- Muller, Nicholas Z. 2014. "Boosting GDP Growth by Accounting for the Environment." *Science* 345 (6199): 873–74.
- Muller, Nicholas. 2020. *AP4 (AP3, AP2, APEEP) Model*. <https://nickmuller.tepper.cmu.edu/APModel.aspx> (accessed July 9, 2020).
- Muller, Nicholas Z., and Robert Mendelsohn. 2009. "Efficient Pollution Regulation: Getting the Prices Right." *American Economic Review* 99 (5): 1714–39.
- Murphy, Kevin M., and Robert H. Topel. 2006. "The Value of Health and Longevity." *Journal of Political Economy* 114 (5): 871–904.
- National Bureau of Economic Research. 2020a. *Mortality Data —Vital Statistics NCHS' Multiple Cause of Death Data, 1959–2017*. <https://data.nber.org/data/vital-statistics-mortality-data-multiple-cause-of-death.html> (accessed April 10, 2020).
- National Bureau of Economic Research. 2020b. *US State and County Population Data by Age, Race, Sex, Hispanic 1969–On. Survey of Epidemiology and End Results*. <https://www.nber.org/research/data/surveyepidemiology-and-end-results-seer-us-state-and-county-population-data-age-race-sex-hi> (accessed April 21, 2020).
- National Research Council. 2006. *State and Federal Standard for Mobile-Source Emissions*. National Academies Press.
- Oates, Wallace E. 1997. "On Environmental Federalism." *Virginia Law Review* 83 (7): 1321–29.
- OECD. 2012. *Mortality Risk Valuation in Environment, Health and Transport Policies*. OECD.
- Office of Management and Budget. 2009. *2009 Report to Congress on the Benefits and Costs of Federal Regulations and Unfunded Mandates on State, Local, and Tribal Entities*. OMB.
- Pigou, A. C. 1932. *The Economics of Welfare*. 4th ed. Macmillan.
- Rucker, S. 2018. *Emission Reduction Credits*. Colorado Air Pollution Control Division.
- Salz, Tobias. 2022. "Intermediation and Competition in Search Markets: An Empirical Case Study." *Journal of Political Economy* 130 (2): 310–45.
- Sanders, Nicholas J., Alan I. Barreca, and Matthew J. Neidell. 2020. "Estimating Causal Effects of Particulate Matter Regulation on Mortality." *Epidemiology* 31 (2): 160–67.
- Shapiro, Joseph S. 2022. "Pollution Trends and US Environmental Policy: Lessons from the Past Half Century." *Review of Environmental Economics and Policy* 16 (1): 42–61.
- Shapiro, Joseph S., and Reed Walker. 2018. "Why Is Pollution from US Manufacturing Declining? The Roles of Environmental Regulation, Productivity, and Trade." *American Economic Review* 108 (12): 3814–54.
- Shapiro, Joseph S., and Reed Walker. 2025. *Data and Code for "Is Air Pollution Regulation Too Lenient? Evidence from US Offset Markets"*. American Economic Association; distributed by Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E203842V>.
- Smeets, Winand, Wim Blom, Anco Hoen, Benno Jimmink, Robert Koelemeijer, Jeroen Peters, and Wilco de Vries. 2007. "Cost-Effective Abatement Options for Improving Air Quality in the Netherlands." Unpublished.
- Smith, V. Kerry, and Ju-Chin Huang. 1995. "Can Markets Value Air Quality? A Meta-Analysis of Hedonic Property Value Models." *Journal of Political Economy* 103 (1): 209–27.
- Sorensen, Alan T. 2000. "Equilibrium Price Dispersion in Retail Markets for Prescription Drugs." *Journal of Political Economy* 108 (4): 833–50.

- Stavins, Robert N. 2003. "Experience with Market-Based Environmental Policy Instruments." In *Handbook of Environmental Economics*, Vol. 1, edited by Karl-Göran Mäler and Jeffrey R. Vincent, 355–435. North Holland.
- Tessum, Christopher W., Jason D. Hill, and Julian D. Marshall. 2017. "InMAP: A Model for Air Pollution Interventions." *PLoS ONE* 12 (4): e0176131.
- Texas Commission on Environmental Quality. 2019a. *Point Source Emissions Inventory*. <https://www.tceq.texas.gov/airquality/point-source/eei/psei.html> (accessed October 18, 2019).
- Texas Commission on Environmental Quality. 2019b. *TCEQ Records Online*. https://records.tceq.texas.gov/cs/idcplg?IdcService=TCEQ_SEARCH (accessed June 16, 2020).
- US Bureau of Labor Statistics. 2020a. *County-Level Population Estimates*. <https://www.bls.gov/cew/downloadable-data-files.htm> (accessed January 26, 2020).
- US Bureau of Labor Statistics. 2020b. *NAICS-Based Data Files. Quarterly Census of Employment and Wages*. <https://www.bls.gov/cew/downloadable-data-files.htm> (accessed January 26, 2020).
- US Bureau of Labor Statistics. 2023. *Survey Output Servlet, Data Access Service*. <https://data.bls.gov/pdq/> (accessed October 9, 2023).
- US Census Bureau. 2010. *Population, Population Change, and Estimated Components of Population Change: April 1, 2010 to July 1, 2019*. https://www.census.gov/data/tables/time-series/demo/popest/2010s-counties-total.html#par_textimage (accessed March 19, 2020).
- US Census Bureau. 2017. *NAICS Concordances*. <https://www.census.gov/eos/www/naics/concordances/concordances.html> (accessed July 14, 2021).
- US Census Bureau. 2019. *Cartographic Boundary Files - Shapefile*. <https://www.census.gov/geographies/mapping-files/time-series/geo/carto-boundary-file.html> (accessed July 14, 2021).
- US Census Bureau. 2020a. *2017 Population Estimates FIPS Codes*. <https://www.census.gov/geographies/reference-files/2017/demo/popest/2017-fips.html> (accessed July 14, 2021).
- US Census Bureau. 2020b. *County Population Totals: 2010–2019*. https://www.census.gov/data/tables/time-series/demo/popest/2010s-countiestotal.html#par_textimage (accessed March 27, 2020).
- US Census Bureau. 2020c. *National Population by Characteristics: 2010–2019*. <https://www.census.gov/data/tables/time-series/demo/popest/2010s-national-detail.html> (accessed March 27, 2020).
- US Environmental Protection Agency. 1980. *Emission Reduction Banking: An Annotated Slide Presentation*. NSCEP.
- US Environmental Protection Agency. 2010. *Conversion Factors for Hydrocarbon Emission Components*. NSCEP.
- US Environmental Protection Agency. 2012. *Regulatory Impact Analysis for the Final Revisions to the National Ambient Air Quality Standards for Particulate Matter*. USEPA.
- US Environmental Protection Agency. 2019a. *Data Downloads (ECHO). Enforcement Compliance History Online*. <https://echo.epa.gov/tools/datadownloads#downloads> (accessed October 18, 2019).
- US Environmental Protection Agency. 2019b. *Green Book Data Download. Nonattainment Areas for Criteria Pollutants*. <https://www.epa.gov/green-book/green-bookdata-download> (accessed December 30, 2019).
- US Environmental Protection Agency. 2020a. *2017 National Emissions Inventory Data. National Emissions Inventory*. ftp://newftp.epa.gov/air/nei/2017/doc/flat_files/ (accessed May 11, 2020).
- US Environmental Protection Agency. 2020b. *Air Emissions Inventories. National Emissions Inventory*. <https://gaftp.epa.gov/Air/nei/> (accessed June 30, 2020).
- US Environmental Protection Agency. 2020c. *Pollutant Emissions Summary Files for Earlier NEIs*. <https://www.epa.gov/airemissions-inventories/pollutant-emissions-summary-files-earlier-neis-> (accessed June 30, 2020).
- US Environmental Protection Agency. 2023. "Mortality Risk Valuation." <https://www.epa.gov/environmental-economics/mortality-risk-valuation> (accessed March 15, 2023).
- van Harmelen, A., H. Kok, and A. Visschedijk. 2001. *Potentials and Costs to Reduce pm10 and pm2.5 Emissions from Industrial Sources in the Netherlands*. Netherlands Organisation for Applied Scientific Research (TNO).
- van Soest, Daan P., John A. List, and Tim Jeppesen. 2006. "Shadow Prices, Environmental Stringency, and International Competitiveness." *European Economic Review* 50 (5): 1151–67.
- Walker, W. Reed. 2013. "The Transitional Costs of Sectoral Reallocation: Evidence from the Clean Air Act and the Workforce." *Quarterly Journal of Economics* 128 (4): 1787–835.
- Woodruff, Tracey J., Jennifer D. Parker, and Kenneth C. Schoendorf. 2006. "Fine Particulate Matter (PM_{2.5}) Air Pollution and Selected Causes of Postneonatal Infant Mortality in California." *Environmental Health Perspectives* 114 (5): 786–90.
- Zhou, Xuehua, Zhaoyu Cao, Yujie Ma, Linpeng Wang, Ruidong Wu, and Wenxing Wang. 2016. "Concentrations, Correlations and Chemical Species of PM_{2.5}/PM₁₀ Based on Published Data in China: Potential Implications for the Revised Particulate Standard." *Chemosphere* 144: 518–26.